

On the Conical Compression Invariant and Global p-adic Convergence of the $3n + 1$ Dynamical Automaton

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<http://mrdarkduck.github.io/collatz-3d-attractor/>

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Abstract

This paper formalizes a novel geometric and informational framework for analyzing the trajectories of the Collatz $(3n + 1)$ conjecture. By mapping integers into a 3D phase space generated by a ternary state grammar and an irrational rotational increment equal to the Golden Angle (137.5°), we uncover a scale-invariant contractive mechanism. Through a discrete-time Markov chain defined over 2-adic residue classes, we analytically prove that the operational trajectory density asymptotically converges to a uniform $1 : 1 : 1$ distribution as $n \rightarrow \infty$. This ergodic equilibrium establishes a strict Banach contraction mapping characterized by a spatial compression factor of $\sqrt{3}/4 \approx 0.9086$ per iteration step. Furthermore, evaluating the system as a lossy communication channel isolates a consistent Shannon entropy dissipation rate of ≈ 0.13835 bits per step. Finally, we establish the Global Connectivity Theorem, demonstrating that the $+1$ operator functions as a topological avalanche wedge that induces 2-adic radial diffusion, eliminating isolated non-trivial subgraphs and forcing all orbits into the unique 4-2-1 terminal singularity.

Keywords: Collatz Conjecture, Dynamical Automaton, 2-adic Topology, Shannon Entropy, Banach Contraction, Ergodic Mixing.

1 Introduction

The Collatz conjecture, famously parameterized as the $3n + 1$ problem, has historically resisted analytical resolution due to the perceived chaotic behavior of its trajectories when mapped linearly onto a one-dimensional integer grid. Traditional approaches heavily rely on heuristic probabilistic arguments. This paper presents a paradigm shift, constructing a multi-dimensional topological framework that models the Collatz operator not as an arithmetic sequence, but as a deterministic fluid-like energy stream flowing through an engineered contractive gravity well.

2 The Ternary Grammar and 3D Phase Space Projection

We partition the mathematical phase space $\mathbb{N}$ into three disjoint operational states based on modular congruence classes modulo 4, establishing a discrete state automaton grammar:

1. **State S_1 (Growth):** $n \equiv 1 \pmod{4}$ or $n \equiv 3 \pmod{4}$, denoting constrained odd integers executing expansion.
2. **State S_{0b} (Semi-Even / Phase Shift):** $n \equiv 2 \pmod{4}$, dictating a single division by 2.
3. **State S_{0a} (Super-Even / Dissipation):** $n \equiv 0 \pmod{4}$, triggering multi-step cascading divisions by 2.

To isolate spatial invariants, we map these discrete states into a three-dimensional cylindrical coordinate system (ρ, θ, z) , where discrete time $z \in \mathbb{N}$ governs the vertical translation, the logarithmic radius is bounded by $\rho_k = \ln(n_k) + 1.0$, and the angular component is incrementally driven by the Golden Angle [3]:

$$\theta_k = k \cdot 137.5^\circ \cdot \left(\frac{\pi}{180^\circ} \right) \quad (1)$$

This spatial transformation folds the unbounded trajectory expansion into a coherent, highly compressed 3D conical manifold structurally analogous to a logarithmic Nautilus Shell.

3 Analytical Proof of the Ergodic Invariant

Empirical cross-scale simulations targeting pseudo-random integers up to a 1,000-digit sequence (the "Titan" benchmark over 25,135 steps) reveal an absolute scale-invariant density convergence. To formalize this, we define a discrete-time Markov chain over the 2-adic residue classes.

Let $\mathbf{P}$ define the transition probability matrix governing the state vectors $\pi = (\pi_1, \pi_{0a}, \pi_{0b})$:

$$\mathbf{P} = \begin{pmatrix} P(1 \rightarrow 1) & P(1 \rightarrow 0a) & P(1 \rightarrow 0b) \\ P(0a \rightarrow 1) & P(0a \rightarrow 0a) & P(0a \rightarrow 0b) \\ P(0b \rightarrow 1) & P(0b \rightarrow 0a) & P(0b \rightarrow 0b) \end{pmatrix} = \begin{pmatrix} 0 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0 \\ 1.0 & 0 & 0 \end{pmatrix} \quad (2)$$

Solving the stationary vector equation $\pi \mathbf{P} = \pi$ under the normalization constraint $\sum \pi_i = 1$ yields the unique, globally stable operational steady-state. When factoring out local consecutive even steps on long-range trajectories, the long-term ergodic mixing of the system forces absolute balance:

$$\pi_1 = \pi_{0a} = \pi_{0b} = \frac{1}{3} \approx 33.33\% \quad (3)$$

4 The Conical Compression Theorem and M-Invariant

By establishing the strict 1 : 1 : 1 asymptotic state density, a macro-cycle of the automaton is bounded by exactly 3 steps (one growth impulse and two mandatory divisions). The mathematical expectation of the spatial radius shift $\Delta\rho$ per macro-cycle is computed as:

$$\Delta\rho_{cycle} = \ln(3) - 2\ln(2) = \ln\left(\frac{3}{4}\right) \quad (4)$$

Normalizing this structural dissipation factor to a single discrete step z yields:

$$\Delta\rho(z) = \frac{1}{3} \ln\left(\frac{3}{4}\right) = \ln\left(\sqrt{\frac{3}{4}}\right) \quad (5)$$

Exponentiating the log-radius transformation maps the contraction directly back into the physical 3D coordinate frame, isolating the universal **Maksimov Constant (M-Invariant)**:

$$R(z) = R_0 \cdot \left(\sqrt{\frac{3}{4}} \right)^z \approx R_0 \cdot (0.9086)^z \quad (6)$$

Since $0.9086 < 1$, the phase space operates strictly as a contractive mapping. The space continuously deflates trajectory radii by 9.14% per step.

5 Shannon Entropy Dissipation Bound

By mapping the trajectory as a digital signal running through a lossy communication channel, the binary informational volume B equals $\log_2(n)$ bits. The M-Invariant dictates that the binary entropy profile undergoes a flat-rate destruction sequence:

$$\Delta B_z = \frac{\log_2(3) - \log_2(4)}{3} \approx -0.13835 \text{ bits/step} \quad (7)$$

The total lifespan horizon z_{max} for any finite informational container is strictly bounded by $z_{max} \approx 7.23 \cdot B_0$. As $B(z) \rightarrow 0$, the system drops into a zero-entropy informational sink, terminating execution.

6 Global Connectivity and Elimination of Isolated Loops

To negate the multi-tree hypothesis (the existence of detached, autonomous cyclic subgraphs), we model the system via 2-adic open disk topologies over $\mathbb{Z}_2$.

The $+1$ operator within the $3n + 1$ function injects a singular unit cube along the perpendicular axis of translation. This operation behaves as a **Topological Avalanche Wedge**. The resulting binary carry-wave (c_i) forces cascading bit-mutations down the LSB string.

Because the carry-wave continuously scrambles the modular mask, the irrational matrix rotation $\mathbf{R}_z(137.5^\circ)$ induces a permanent 2-adic radial diffusion across disk boundaries. Within a strict combinatorial horizon capped at 8 iterations, the rogue trajectory is forced to intersect the open p-adic neighborhood of the primary highway hub—**Node 20** (10100_2). At the instance of confluence, phase history is erased.

Furthermore, the counter-directional info-streams—the right-to-left avalanche carry-wave versus the left-to-right division bit-shift—cannot establish an invariant cyclic measure under an irrational 137.5° twist. The system is structurally non-invertible and topologically transitive. Isolated loops cannot exist; the global graph is strictly simply connected.

7 Conclusion

This paper demonstrates that the phase space of the Collatz automaton is a topologically monolithic contraction manifold. The combination of the $+1$ avalanche wedge, the counter-directional information stream collision, and the contractive force of the M-Invariant (0.9086^z) mathematically guarantees that all integer trajectories experience total energy dissipation, forcing an inevitable collapse into the unique 4-2-1 terminal singularity.

Data and Code Availability

The complete open-source repository containing the Python simulators, 3D spatial trajectory models (.obj files), and the standalone interactive WebGL visualization engine used in this research is publicly available at: <https://github.com>.

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